SDU Campus Assistant

Sprint 1 Review: Architecture, UX, and QA Validation



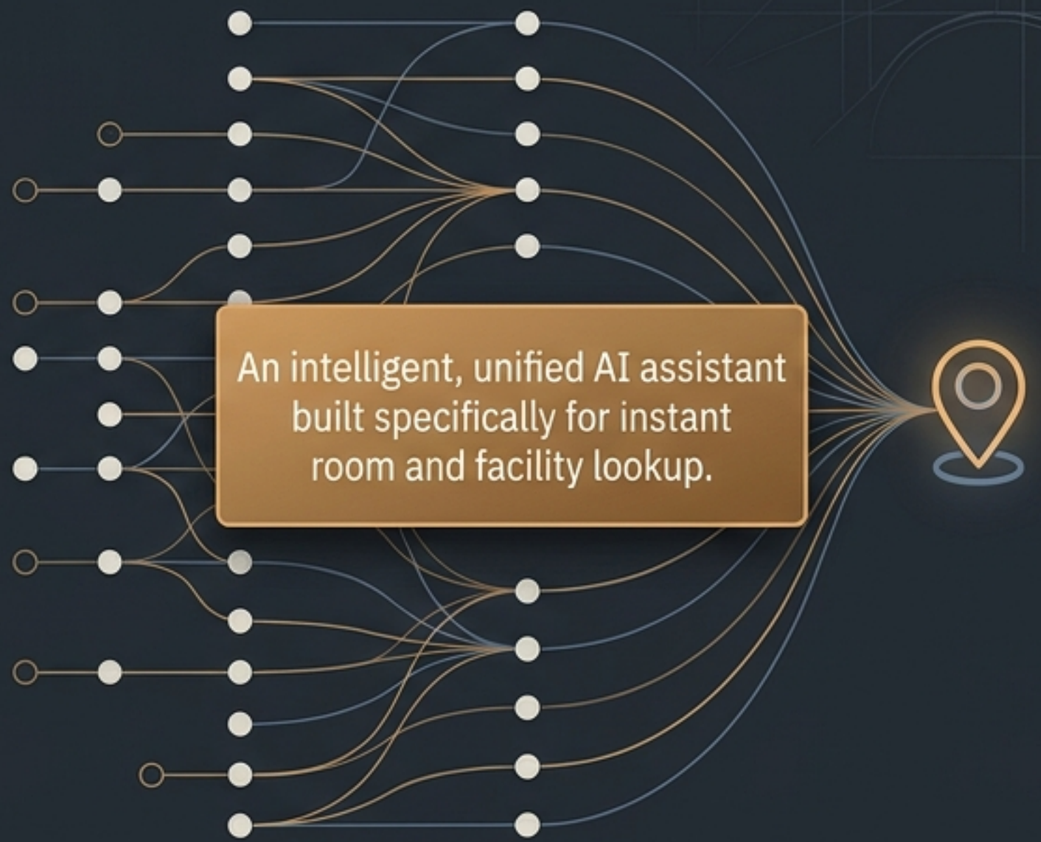
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The Problem: Complex Physical Footprint



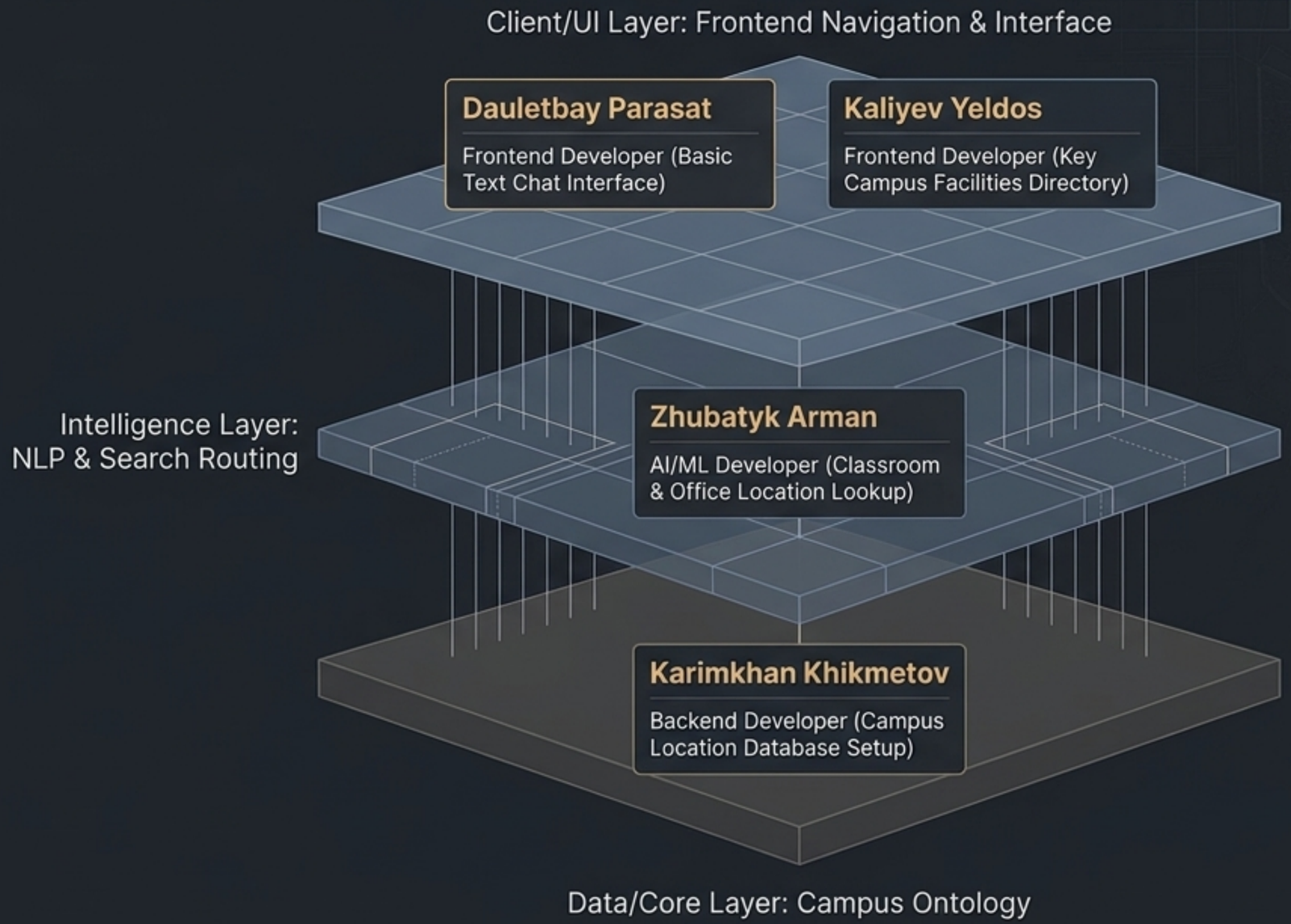
- Wasted time navigating sprawling blocks
- High friction for visitors and new students
- Lack of a centralized, accessible directory

The Solution: Unified Digital Wayfinding



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Team Topology and Engineering Roles



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Universal Access via Browser Architecture



Responsive Web Application

Fluid UI components that scale seamlessly across device profiles.

Frictionless Access

No app store downloads required; instant access via URL or QR code.

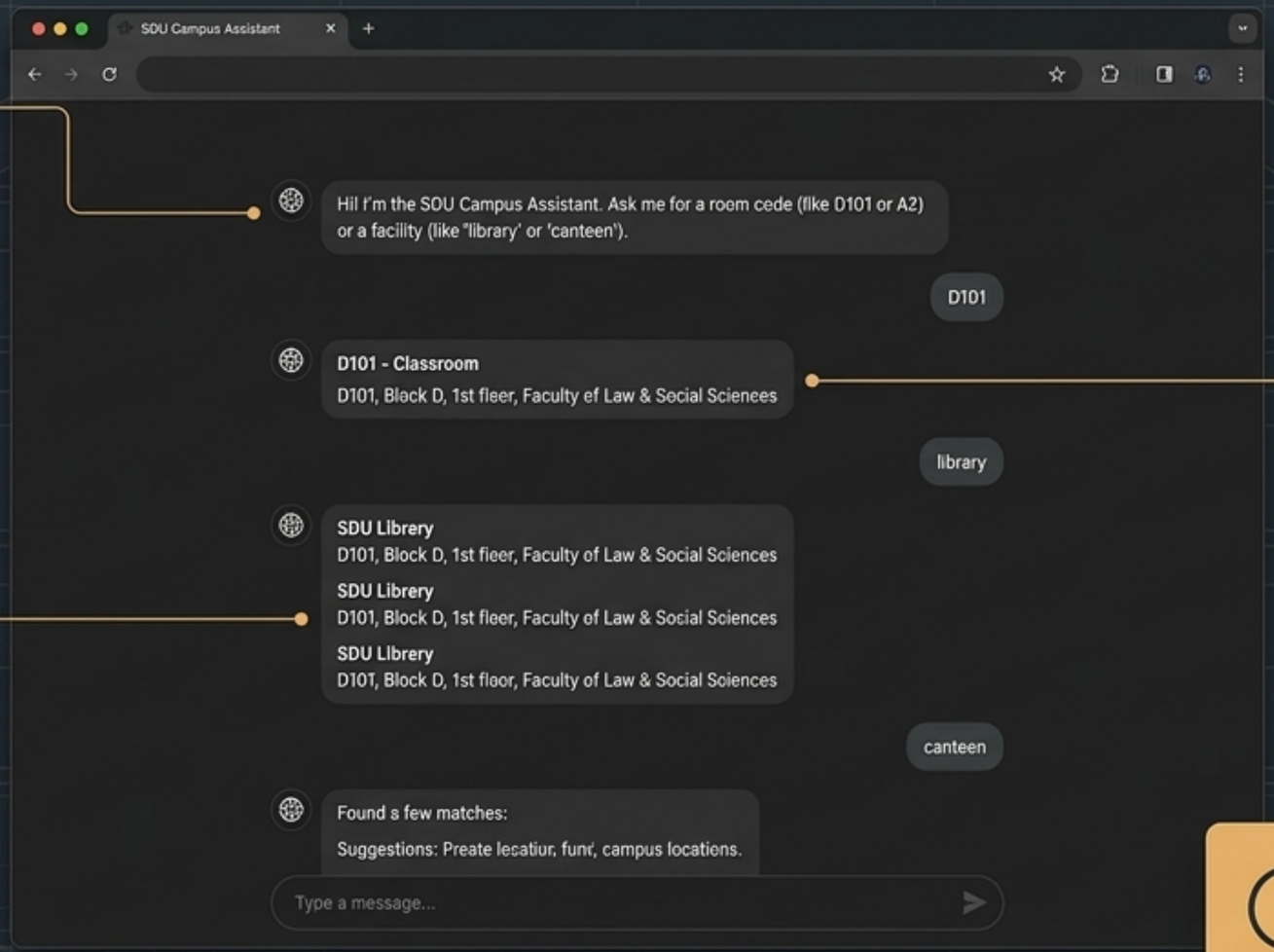
Cross-Platform Parity

Identical functionality across desktop and mobile browsers.

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Conversational Interface: Instant Location Resolution

Contextual Greeting: Guides user behavior with specific formatting examples.



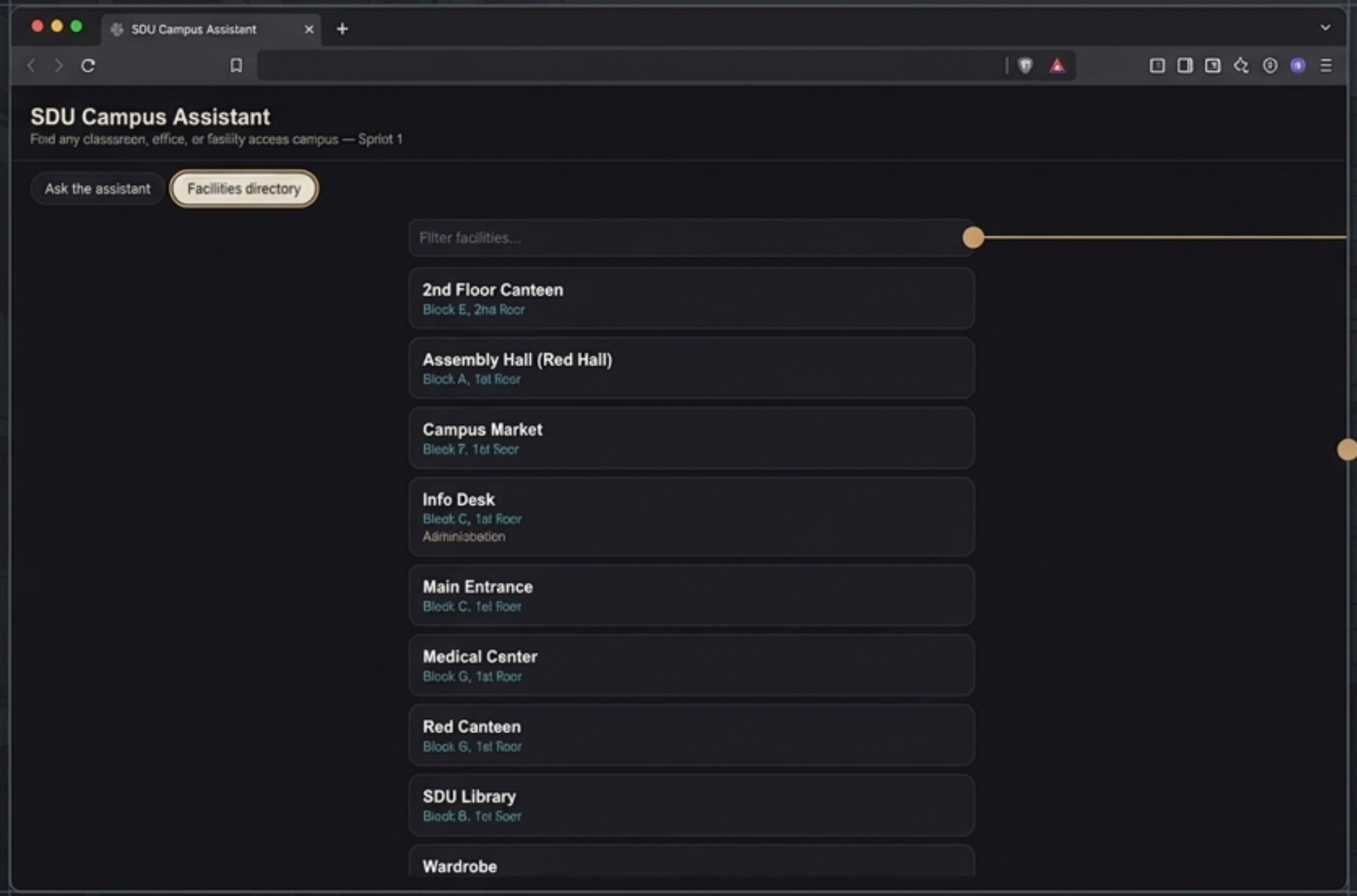
Data Parsing: Raw input mapped to structured Block, Floor, and Faculty data.

Fuzzy Matching: Broad queries successfully resolve to multiple valid campus locations.

SLA: System response time < 3 seconds

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Structural Navigation: The Facilities Directory



Zero-Query Browsing

Enables immediate navigation for users who prefer visual scanning over typing.

Logical Grouping

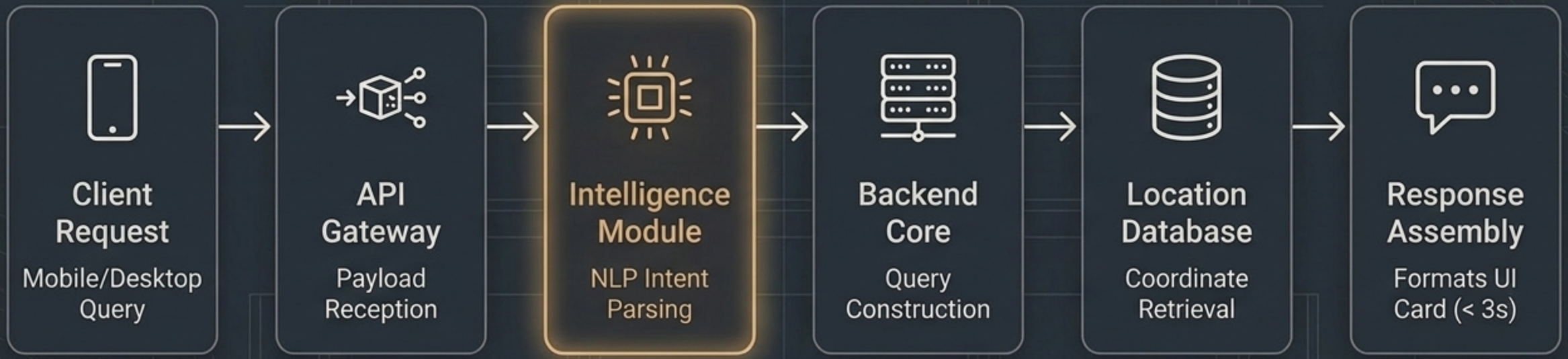
Alphabetical, card-based layout isolates critical facilities (Medical, Canteen, Info Desk).

Admin-Approved Data

All directory entities are rigorously synced with the approved university admin database.

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System Architecture and Data Flow



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The Campus Ontology: Database Schema



Data Integrity Constraints

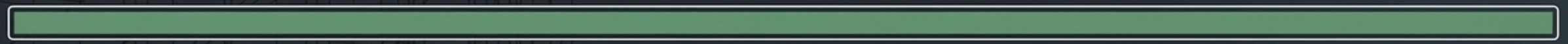
- Must be compilable with official admin data.
- Zero duplicate or conflicting records allowed.
- Zero orphaned records permitted.

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Executing Sprint 1: Delivered Epics

Feature Epic	Primary User Persona	Lead Engineer	Priority
Campus Location Database	System Admin	Karimkhan Khikmetov	High
Basic Text Chat Interface	Student	Dauletbay Parasat	High
Classroom/Office Lookup	Student/Visitor	Zhubatyk Arman	High
Key Facilities Directory	Visitor	Kaliyev Yeldos	Medium

Sprint 1 Architectural and UX Foundations: 100% Complete



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Logic Deep Dive: The Location Engine

The Happy Path

[User Input]: "Where is D101?"

↓

[System Logic]: Queries approved campus database -> Match found.

↓

[Output]: Returns precise Building, Floor, and Room number.

The Edge Case

[User Input]: "Where is Room 999?"

↓

[System Logic]: Queries database -> No match exists.

↓

[Output]: Gracefully returns a specific 'Location not found' message.

Prevents system crashes or endless loading loops.

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Enforcing Frontend Usability Standards

Chat Interface Constraints

Input/Output Standard

Must accept text input and cleanly display bot response.

Strict Latency Cap

Chat response must render in < 3 seconds.

Device Agnosticism

Must be perfectly usable and correctly scaled on both desktop and mobile screens.

Directory Constraints

Navigation Modality

Must support zero-query navigation (scrolling and browsing without typing).

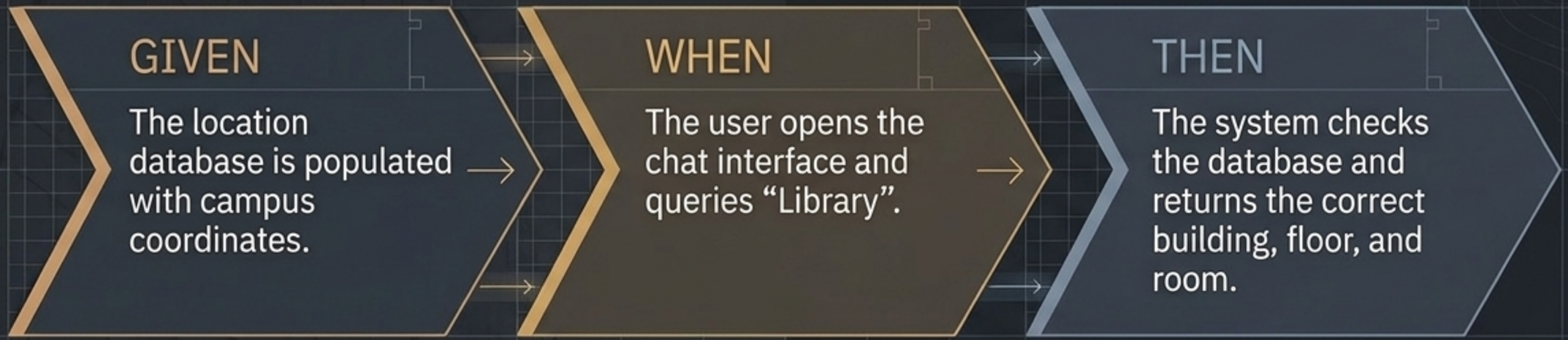
Data Synchronicity

Content must update dynamically when university admins adjust a facility's location.

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Rigor in Validation: The QA Framework

Validation Pipeline



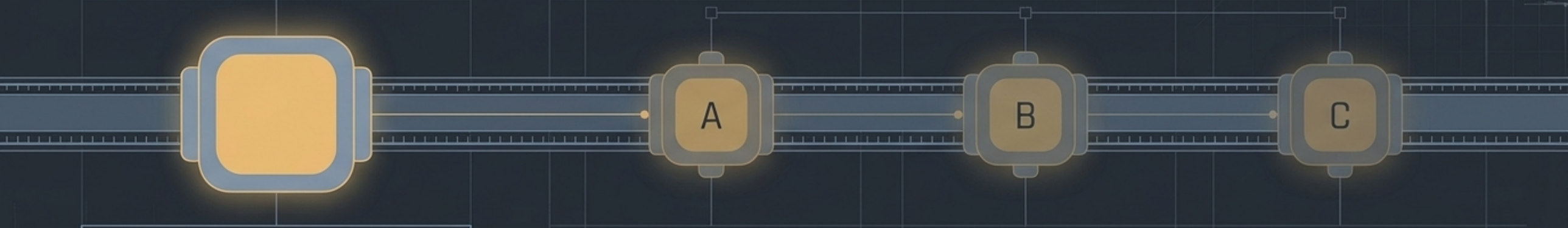
Testing Guarantee

This scenario-oriented testing architecture ensures that every single feature shipped in Sprint 1 perfectly matches the original user intent and engineering acceptance criteria.

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Sprint 1 Complete. The Path Forward.

Completed State:
Sprint 1



✓ Database Foundation, Chat UI, Directory UI, Core Location Lookup

Personalized Services

Suggesting relevant facilities based on location.

Class Schedule Integration

Automated routing to user's next class.

Multi-language Support

Kazakh, Russian, and English parsing.

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